

ODAH EMMANUEL OKOLI

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RESEARCH INTEREST

Passionate about animal genetics, genomics, statistical modeling of host-microbiome interactions, digital phenotyping, and microbial gene association to improve phenotypes, animal welfare, and sustainable production systems.

EDUCATIONAL BACKGROUND

April 2025 Present.	PhD Candidate in Animal Genetics and Genomics <i>Norwegian University of Life Sciences (NMBU), Norway.</i> Project: Hologenomics of Resource Efficiency in Atlantic Salmon. Doctoral Network: Horizon Europe Marie Skłodowska-Curie Action (HoloGen) Focus: Interaction between Host genome and gut microbiome affecting omega-3 fatty acid content and feed efficiency in Atlantic salmon. Methods: Genomic prediction, Microbiome and Microbiability analysis, Digital Phenotyping, Multi-omics integration, Statistical modelling, and Microbial gene association with feed efficiency and omega-3 content.
Aug. 2023 - June 2024.	AgroParisTech, University Paris Saclay, France. Master of Science (European Master in Animal Breeding and Genetics - EMABG - Joint Master) <i>Specialization: Predictive and Integrative Animal Biology.</i> CGPA: B (French Grading System), 70.5% .
Aug. 7, 2023- Aug. 18, 2023.	University of Natural Resources and Life Sciences, Vienna (BOKU), Austria. <i>Specialization: Contribution of Animal Breeding to Global Food Security.</i> Euroleague for Life Sciences Summer School. Grade: 2.0 (B).
Sept. 2022 – June 2023.	Georg August University of Göttingen (UGOE), Germany. Master of Science (European Master in Animal Breeding and Genetics - EMABG - Joint Master) <i>Specialization: Integrated Plant and Animal Breeding.</i>
Aug. 2019 - Oct. 2023.	Federal University of Agriculture Abeokuta, Nigeria. Master of Agriculture. <i>Specialization: Animal Breeding and Genetics.</i> Distinction: 75.38% , Ranking: Top 5%.
Oct. 2011 - Oct. 2017.	Federal University Wukari, Taraba State, Nigeria. Bachelor of Agriculture. <i>Specialization: Animal Science (First Class Hons.)</i> CGPA: 4.61/5.00 (92.2%), Ranking: Top 5%.

WORK EXPERIENCE

Jan. 2024 - June 2024.	GRADUATE STUDENT RESEARCHER (Master Thesis). Bovine Genetics and Genomics Team, National Research Institute for Agriculture, Food, and Environment (INRAE), France. • Conducted statistical pretreatments on 36,986 Mid-infrared (MIR) spectra of milk from three French dairy cattle breeds. • Using processed spectra and SNP-genotypes, I estimated genetic parameters (heritability of absorbances, genetic variance), and assessed the ability of MIR spectra to explain genetic variability. • Predicted breeding values using 50K SNP-genotype and MIR spectra data for milk production (protein and fat yields), health (somatic cell count and mastitis), and fertility traits (heifer and cow conception rates). • Compared the accuracies of Genomic (GBLUP) and Phenomic (spectra-based) prediction models. • Integrated Genomic and Phenomic data to predict breeding values using the BGLR platform, and estimated prediction accuracy.
March 1, 2023- March 31, 2023.	GRADUATE RESEARCH INTERN. Institute for Farm Animal Genetics - Friedrich-Loeffler-Institute (FLI), Germany. • Genotype Data Conversion - I converted genotype call codes of different array versions from Affymetrix IMAGE coding system to a reference-based coding system. • Genotype Imputation: Performed genotype imputation for a lower-density dataset (10K SNPs) genotyped with Affymetrix IMAGE, using a higher-density reference dataset (600K SNPs) genotyped with the Affymetrix Axiom Genome-Wide chicken array. • Assessment of Genotype Imputation Accuracy- I performed 5-fold cross validation to and estimated the accuracy of imputed genotype information.
July 2020 - December 2020.	GRADUATE RESEARCH INTERN. FUNAAB Alpha Poultry Breeding Farm, University of Agriculture Abeokuta. • Performed semen collection, artificial insemination (AI), chick-sex determination, and sorting of sexed chicks by major gene indicators. • Assisted in designing and monitoring of chicken experimental project, and co-supervised five undergraduate interns. • Achievements: Generated 1,325 birds in 3 batches via AI and differentiated chick sex at day-old using vent technique.
Nov. 2018 - Oct 2019.	GRADUATE RESEARCH AND TEACHING ASSISTANT. Department of Animal Science, Federal University Wukari, Taraba State, Nigeria. • Phenotype and Sample collection: Conducted morphometric and linear body measurements on sheep, goats, and poultry. Collected (semen, fecal, and blood) samples for use in AI, hematology, DNA assay, and phenotypic characterization. • Educational Support: Assisted in organizing tutorials and teaching undergraduate students basic Biostatistics, Principles of Animal Husbandry, and Ethics of scientific report writing.

Nov. 2017- Oct 2018.	ANIMAL HUSBANDRY TEACHER. Amazing Grace International College Jos, Plateau State, Nigeria. - Taught <i>"Basic Animal Husbandry Techniques and Practice"</i> to 45 senior secondary students and <i>"Agricultural Science"</i> to 200 junior secondary students, achieving an average success rate of 69.4% for senior students and 73.5% for junior students. - Administered examinations and prepared progression reports for 245 students.
March 2016- Sept. 2016.	BACHELOR (BSc) INTERNSHIP. National Animal Production Research Institute (NAPRI), Zaria Kaduna State, Nigeria. I gained practical experience in animal biotechnology techniques, oestrous detection, and artificial insemination in cows, semen collection from Holstein Friesian crosses and <i>Bunaji</i> bulls, semen evaluation, and farm animal handling.
SCIENTIFIC RESEARCH EXPERIENCE	
Jan. 2024- June 2024.	Master's Thesis (Joint master's degree). DEEP PHENOMIC PROJECT. Evaluating the Efficiency of Phenomic Selection in Dairy Cattle. <i>Supervisors: Dr. Pascal Croiseau, Dr. Cuyabano, B., Dr. Sanchez, M.P (INRAE, France), and Prof. Dr. Armin Schmitt (University of Gottingen, Germany).</i> Achievement: This research project has been accepted for presentation at the upcoming European Federation of Animal Science conference (EAAP) in Florence, Italy, in September 2024.
Nov. 2023- Dec. 2023.	M2 Scientific Research project (University Paris Saclay, France). Characterization of Blood Cell Profile in piglets Weaned at Different ages. <i>Supervisors: Dr. Fanny Blanc and Dr. Jordi Estellé (INRAE, Jouy-en Josas, France).</i>
1 March- 30 th March 2023.	Graduate Internship. SYNBREED Chicken Diversity Project. Matching of Genotype information of different arrays after conversion by means of imputing. <i>Supervisors: Dr. Christian Reimer and Prof. Dr. Steffen Weigend (FLI, Germany).</i>
February 2019.	Master's Thesis FUNAAB Alpha indigenous Chicken Improvement Project. Molecular and Morphometric Characterization of Eimeria Parasites in Chickens. - Extracted 100 Eimeria genomic DNA in faecal samples from six chicken genotypes. Achievement: Identified 7 <i>Eimeria spp</i> in chickens using microscopy and PCR methods, defined their morphometric and molecular profiles, established spp prevalence across host genotypes and production systems, and inferred their evolutionary relationships.
April 2017 - Aug. 2017	Bachelor's Thesis (B. Agric.) Phenotypic Characterization of Local Chickens (<i>Gallus gallus domesticus</i>). - I examined 530 adult birds and 111 eggs from 7 distinct in Bekwarra, Nigeria, for phenotypic characterization. Achievement: Provided the first comprehensive phenotypic characterization data, which currently forms a baseline reference for chicken populations in Bekwarra.
2018	Qualitative Traits Variation in Chickens. Collaborated with Professor Daikwo's Lab at Federal University Wukari, and we determined the <i>Distribution and Gene frequency of Qualitative traits</i> in 1060 chickens and established variation in gene segregation and further characterized the traits.

WORKSHOPS AND PROFESSIONAL TRAINING

BASIC GENETICS AND GENOMIC ANALYSIS

5 -11
November 2023. *Genetics and Genomics unit, INRAE and PRIAM, AgroParisTech, France.*
Curriculum: Using NCBI Databases • Cattle Atlas • Association studies • QTL detection to Gene identification • Introduction to Genomic Evaluation with GBLUP and Bayesian Alphabets.

ADVANCED GENETICS COURSE

4 - 10
December 2023. *Genetics and Genomics unit, INRAE and PRIAM, AgroParisTech, France.*
Curriculum: Linear Mixed Model • Genetic Evaluation and BLUP • Animal and Sire model • Genetic Evaluation and BLUP • Prediction of Breeding Values using GBLUP and RKHS methods.

Hands-on- Experience in Molecular Biology and Bioinformatics Workshop

15-18
November 2021. *African Biosciences Training Institute, Ibadan and Centre for Biotechnology, University of Agriculture, Abeokuta.*
Curriculum: Practical DNA Extraction, Quantification • Next Generation Sequencing principles • Sequence Alignment • Phylogenetics • Marker-Assisted Selection in Animals • PCR Procedures, Electrophoresis and Gel documentation.

DIGITAL SKILLS

R - Programming	• Data visualisation	• SAS	• Plink	• BGLR	• MEGA
BEAGLE	• CLUSTALW	• MS-Excel	• MS-Word	• PowerPoint presentation.	

SCIENTIFIC PUBLICATIONS

Odah, E.O., Wheto, M., Ojaowo H., Ogunlola, O. O., Adebambo, A.O. and Adebambo, O. **Relationship between Body weight and Semen quality traits of Pure and improved indigenous Nigerian cocks.** 2022. Conference Proceedings of the Nigerian Society for Animal Production held at the University of Jos, March 2022, pages 181–185. <https://www.researchgate.net/publication/359443082>

Odah, E. O., Daikwo, S. I and Oko, B. O. 2019. **The Role of Biotechnology in Animal Agriculture in an era of Climate change.** *International Journal of Animal Science, Husbandry and Livestock Production*, ISSN: 2141-5191 Volume. 5 (4), page 272-276, September 2019. Available online at <http://www.advancedscholarsjournals.org>

Odah, E. O., Daikwo, S. I., Mbap S.T. and Okpanachi U. 2019. **Phenotypic Characterization of Local chickens (*Gallus gallus domesticus*) in Bekwarra Local Government Area, Nigeria.** *JSM Veterinary Medicine and Research*. <https://www.jsmcenral.org/VeterinaryMedicine/jsmvmr835561.php>

Daikwo, S. I., **Odah, E. O.**, Ogah, D. M. and E. B. T. Baba-onoja. 2018. **Qualitative Traits Variation in Indigenous Chickens of Bekwarra, Nigeria.** *Asian Research Journal of Agriculture*, 2456-561X, Vol.: 9, Issue. <http://www.sciencedomain.org/abstract/24799>

- Other scientific publications to my credit in International and local peer-reviewed journals are available via this link; <https://www.scihorizon.com/RID-20211826>.

SCIENTIFIC CONFERENCES ATTENDED AND PRESENTATIONS

5 -12th May 2024. 2nd International Dairy Cattle Conference held at the Sari University of Iran, Iran.
Virtual presentation: *Potentials of Phenomics Selection in Dairy Cattle: An alternative to Genomic Selection.*

31st May 2024.	R2D2 Annual Conference, INRAE, France. Virtual presentation: <i>Evaluating the Efficiency of Phenomic Selection in Dairy Cattle.</i>
Oct 13–14, 2022.	4 th CIBREED Fall Conference at the Centre for Integrated Plant and Animal Breeding Research, University of Gottingen, Germany. Theme: "Hybrid Breeding and Heterosis", "Innovative Species", and "Innovative Technologies". Poster presented "Comparative Growth Performance of Two Chicken Breeds (Kuroiler and Nigerian indigenous Naked Neck) Raised in a Tropical Environment".
Oct 3 - 5, 2022.	VII. International Congress on Domestic Animal Breeding Genetics and Husbandry, at Samsun, TÜRKİYE. Paper presented: "Variation in growth performance and feed conversion ratio in selected FUNAAB alpha (dual purpose and layer line) chickens".

AWARDS AND HONOURS

March 2022	MacDonald and Cheryl Wick's Prize for the best-graduating student in Animal Genomics and Advanced Biotechnology (Postgraduate) for the 2019-2020 Academic session at the Federal University of Agriculture, Abeokuta, Nigeria.
Oct. 2017	Awarded Best Graduating Student of Animal Science Department of Animal Production and Health Federal University Wukari, Nigeria.
Oct. 2017	Award of Academic Excellence for the Overall Best Graduating Student of Agriculture Faculty of Agriculture and Life sciences, Federal University Wukari, Nigeria.
Jan. 2015	Who's Who Among Students in Nigerian Universities Award Award of Distinguished Academic Excellence and Service by Idyllic Global Links Limited.

SCHOLARSHIP AND FUNDING

2022	Erasmus Mundus Joint Master's Degree Scholarship for the European Master in Animal Breeding and Genetics (EMABG). Project Number: 598102-EPP-1-2018-1-AT-EPPKA1-JMD-MOB. Study host institutions: University of Natural Resources and Life Sciences, Austria, University of Göttingen, Germany and University of Paris, Saclay, France.
2021	United Nations Educational, Scientific, and Cultural Organization UNESCO (master's one-time scholarship). Under the auspices of the International Centre for Biotechnology, University of Nigeria Nsuka. Study host institution: University of Agriculture, Abeokuta.
2020	Tertiary Education Trust Fund (TETFUND) Fellowship for the Animal Breeding and Genetics MSc program. Awarded by Federal University Wukari. Study host institution: Federal University of Agriculture, Abeokuta, Nigeria.

PROFESSIONAL MEMBERSHIP

Member: European Network on Livestock Phenomics (**EU-LI-PHE**)
Working Group (WG1) - Phenotyping Technologies, COST Action CA22112.
Member: European Federation of Animal Science (**EAAP**)
Member: Genetics Society of America (**GSA**)
Member: Animal Science Association of Nigeria (**ASAN**)
Member: Animal Welfare Group, Nigeria (**AWGN**)

VOLUNTEER ACTIVITIES AND LEADERSHIP QUALITIES

Aug. 2022- June 2024.	University of Göttingen (EMABG -Erasmus) Country Representative • Elected country representative of my EMABG study cohort at the University of Göttingen. • Served as the liaison between international students and the program administration. • Ensured students participation and representation in academic and student-related activities within Germany, France, and Austria, and relate students' concerns to administrators.
Jan - December 2016.	President of National Animal Science Students Association (NASSA) Federal University Wukari Branch, Nigeria. • Led a team of 5 elected executives and coordinated 216 members. • Organized seminars and invited experienced professionals in Animal Science to guide students on early career choices, professional development, and industry opportunities for life after graduation. • Successfully registered and incorporated the local association with the National arm.

REFERENCES

Available on Request.